

Riemann-Stieltjes Integral

Let f and g be bounded functions on the real interval $[a, b]$. Let P be a partition of $[a, b]$ of the form $a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$ for some positive integer n . Let $\max_i |x_i - x_{i-1}|$ be the *mesh* of the partition P . The mesh is the length of longest partition element of P . Define

$$S(f, g, P) = \sum_{i=1}^n f(\xi_i)[g(x_i) - g(x_{i-1})] \quad (1)$$

where $\xi_i \in [x_{i-1}, x_i]$. Note that if g is the identity function, the sum collapses to the definition of a Riemann sum; it amounts to using a uniform distribution to measure the “mass” of points. The Stieltjes integral allows for non-uniform distributions on $[a, b]$.

The Stieltjes integral of f with respect to g is defined as the limit (if it exists) of $S(f, g, P)$ as the mesh of the partition tends to 0. Formally, there is some number L such that for every $\varepsilon > 0$ there is a $\delta > 0$ such that for any partition P with mesh less than δ and for any choice of the ξ 's in that partition it follows that

$$|S(f, g, P) - L| < \varepsilon.$$

If the limit exists, then we write

$$L = \int_a^b f(x)dg(x)$$

or simply $\int f dg$. In our application, f is a continuous vN-M utility and g is a cdf with bounded support. In this case, the Stieltjes integral always exists. Sometimes we will integrate the composition of a vN-M utility u with a function ψ that is not continuous. If the composition $f = u \circ \psi$ is bounded and g is continuous at any point of discontinuity of f , then the integral exists. If g is C^1 , then it turns out that we can write the Stieltjes integral of f as a Riemann integral, $\int_{[a,b]} f(x)g'(x)dx$. Rewrite $S(f, g, P)$ as

$$S(f, g, P) = \sum_{i=1}^n f(\xi_i) \left[\frac{g(x_i) - g(x_{i-1})}{x_i - x_{i-1}} \right] (x_i - x_{i-1}).$$

Intuitively, as the mesh tends to 0, the expression in brackets converges to g' , so the limit is the Riemann integral of fg' . In our application if the cdf F is C^1 then we can write $\int u(z)dF(z) = \int u(z)F'(z)dz$, where the derivative F' is often called the density function of the distribution.